6	(a)	(i)	Increasing R increases the total resistance of the circuit, thus reducing the current in the circuit.	[1]
			Thus the potential drop across the internal resistance is reduced.	[1]
			Since the terminal p.d. is the e.m.f. of the battery minus the p.d. across the internal resistance, terminal p.d. increases with R.	[1]
			OR	
			Increasing R increases the external resistance.	[1]
			This increases the ratio of the external resistance to the total resistance.	[1]
			By the potential divider principle, the terminal p.d. increases with R.	[1]
	(a)	(ii)	$I = \dfrac{25.2}{2.00 + 4.00 + 40.0} = 0.548 \text{ A}$	
			$V_B = 25.2 - 0.548(2.00)$	[1]
			$= 24.1 \text{ V}$	[1]
			OR	
			$V_B = \dfrac{4.00 + 40.0}{2.00 + 4.00 + 40.0} \times 25.2$	[1]
			$= 24.1 \text{V}$	[1]
	(b)	(i)	$V_R = (1.60 \times 10^{-3})(1.50 \times 10^{3}) = 2.40 \text{ V}$	[1]
			$V_{thermistor} = 6.00 - 2.40 = 3.60 \text{ V}$ and $R_{thermistor} = \dfrac{3.60}{1.60 \times 10^{-3}} = 2250\,\Omega$	[1]
			From the graph, $t = 14°\text{C}$	[1]
	(b)	(ii)	From the graph, $R_{thermistor} = 3100\ \Omega$ at $t = 5°\text{C}$	[1]
			Based on potential divider principle, potential difference across the fixed resistor	
			$V_R = \left(\dfrac{1500}{1500 + 3100}\right)6.00 = 1.96 \quad \text{V}$	[1]